

SmartCare v0.2 - Requirements Specification Template

1. Problem and Scope

The problem is that SmartCare relies on spreadsheets and paper records, causing duplicate bookings, difficulty finding patient information, inconsistent appointment statuses, and limited history. The main scope is a small, maintainable software system for managing patients, practitioners, and appointments. Integrations like payments or SMS are out of scope. Self-service booking is provisional.

2. Stakeholders

Stakeholder	Need	Evidence
Staff	Fix for duplicate bookings and find information	Staff report duplicate bookings, difficulty finding patient information
Practitioners	Reliable appointment history	Staff report inconsistent appointment status and limited appointment history
Patients	Reliable booking system	The system must be a patient and appointment process
IT Support	Low maintenance	Based on management's requirement for a system that is easy to maintain
Management	Small, maintainable system	Management wants a small, maintainable system

3. Functional Requirements

FR-01: The system should allow staff to create and save a new patient record.

FR-02: The system should allow staff to search for an existing patient by name or ID.

FR-03: The system should allow staff to create an appointment for a patient with a specific practitioner.

FR-04: The system should prevent the creation of overlapping (duplicate) bookings for a single practitioner.

FR-05: The system should allow staff to update an appointment's status for e.g., scheduled, completed, cancelled.

FR-06: The system should retain and display a historical log of all appointments for a patient.

FR-07: The system should allow staff to create and edit practitioner profiles.

FR-08: The system should display a daily schedule of appointments for a selected practitioner.

FR-09: The system should allow staff to edit existing patient contact information.

FR-10: The system should explicitly label cancelled appointments in the history rather than deleting them.

FR-11: The system should allow staff to view available appointment slots for a given date.

FR-12: The system should allow staff to search for appointments by date range.

4. Non-Functional Requirements

NFR-01: The system's architecture must remain small and maintainable, requiring less than maybe an average of 2 hours of IT administration per month.

NFR-02: The system should return patient search results within 2 seconds to resolve difficulty finding information.

NFR-03: The system should ensure appointment status data is consistent and updates in real-time across all staff terminals.

NFR-04: Core business logic for scheduling must be independently testable.

NFR-05: The system database should securely back up patient records daily to prevent data loss.

NFR-06: The system user interface must be intuitive enough that new staff require less than 1 hour of training to book an appointment and/or other necessary features.

5. User Stories

US-01: As a receptionist, I want to search for a patient, so that I can quickly find their information.

US-02: As a receptionist, I want the system to warn me of conflicts, so that I avoid duplicate bookings.

US-03: As a practitioner, I want to view a patient's appointment history, so that I can review their past visits.

US-04: As a receptionist, I want to update an appointment's status, so that the clinic has an accurate schedule.

US-05: As a part of management, I want a simplified user interface, so that the system remains small and easy to maintain.

US-06: As a receptionist, I want to maintain cancelled appointments in the log, so that we have a complete historical record.

6. Acceptance Criteria

GIVEN a patient named "Jane Doe" is saved in the database

WHEN a staff member searches for "Jane Doe"

THEN the system returns Jane Doe's patient profile.

GIVEN an existing appointment for "John Smith"

WHEN a staff member updates the status to "Cancelled"

THEN the system displays the status as "Cancelled" and retains it in the history log.

GIVEN Dr Smith has a booked appointment at 10:00 AM

WHEN staff attempt to book a new appointment for Dr Smith at 10:00 AM

THEN the system blocks the action and displays a duplicate booking error.

7. Assumptions and Open Questions

Assumption: Patients will not have self-service online booking access in v1.0.

Open Question: Does the system need to accommodate multiple clinic locations, or just one single facility?

8. AI Requirements Review Record

AI suggestion	Evidence?	Decision	Reason	Verification
Add SMS reminders	No	Rejected	Client requested a small system; SMS is out of scope	Checked with client brief
Block duplicate bookings	Yes	Accepted	Brief explicitly states staff struggle with duplicate bookings	Checked with client brief
Add facial recognition	No	Rejected	Contradicts management's need for a maintainable, small system	Checked with client brief
Appointment history log	Yes	Accepted	Brief notes current system has "limited appointment history"	Checked with client brief
Online payments	No	Rejected	Billing is not mentioned; system is strictly for patients, practitioners, appointments	Checked with client brief